Supplementary Material for "Graph Spatio-Temporal Alignment for Trajectories of Directed, Partially Observed Attributed Networks"

This supplement contains the proofs (Section S1), algorithmic details (Section S2), additional verification, convergence-audit and benchmark results (Section S3), simulation designs and additional results (Section S4), data and preprocessing (Section S5), additional empirical results (Section S6) proofs and results for the sensitivity analysis (Section S8), and a guide to the reproducibility package (Section S7). Numbered results without the prefix S refer to the main text.

S1 Proofs

Throughout, $G = (\mu, F, C, W)$ on n nodes, $H = (\nu, G, D, W')$ on m nodes, $r = P\mathbf{1}$, $c = P^\top \mathbf{1}$, and $L_{ijkl} = w_{ik} w'_{jl} (c_{ik} - d_{jl})^2$, so that $Q(P) = \sum_{ijkl} L_{ijkl} p_{ij} p_{kl}$.

Proof of Proposition 1. (a) Each term of J_0 is non-negative. J_0 is continuous on $\mathbb{R}_+^{n \times m}$ (with $0 \log 0 = 0$) and coercive because $\mathrm{KL}(r|\mu) \geq \sum_i \{r_i \log(r_i/\mu_i) - r_i\}$ grows superlinearly in $\|P\|_1$; sublevel sets are compact and the minimum is attained. (b) For $P \in \mathbb{R}_+^{n \times m}$, $J_0^{HG}(P^\top) = J_0^{GH}(P)$: the feature term is invariant because m is symmetric; $Q^{HG}(P^\top) = \sum w'_{jl} w_{ik} (d_{jl} - c_{ik})^2 p_{ij} p_{kl} = Q^{GH}(P)$; and the two KL terms are exchanged. Transposition is a bijection between the feasible sets, and E is transposition-invariant, so D_0 and V_ε are symmetric. (c) $P = \mathrm{diag}(\mu)$ gives $J_0 = 0$. (d) If σ exists, the coupling $p_{i\sigma(i)} = \mu_i$ gives $J_0 = 0$. Conversely, at a minimizer with $J_0(P) = 0$ the KL terms vanish, so $P\mathbf{1} = \mu$ and $P^\top \mathbf{1} = \nu$; since $\alpha < 1$ the feature term vanishes, so $p_{ij} > 0$ only if $f_i = g_j$. Distinct attributes imply that each row and each column contains at most one positive entry; positive row and column sums then force a bijection σ with $p_{i\sigma(i)} = \mu_i = \nu_{\sigma(i)}$ and $g_{\sigma(i)} = f_i$, so $n = m$. Since $\alpha > 0$ and all weights are positive, $Q(P) = \sum_{i,k} w_{ik} w'_{\sigma(i)\sigma(k)} (c_{ik} - d_{\sigma(i)\sigma(k)})^2 \mu_i \mu_k = 0$ forces $d_{\sigma(i)\sigma(k)} = c_{ik}$. □

Proof of Corollary 1. The argument for the KL and feature terms in the proof of Proposition 1(d) does not use the weights, so $J_0(P) = 0$ again forces P to be a permutation coupling with $\nu_{\sigma(i)} = \mu_i$ and $g_{\sigma(i)} = f_i$. For such P, $Q(P) = \sum_{i,k} w_{ik} w'_{\sigma(i)\sigma(k)} (c_{ik} - d_{\sigma(i)\sigma(k)})^2 \mu_i \mu_k$

vanishes if and only if $c_{ik} = d_{\sigma(i)\sigma(k)}$ whenever $w_{ik}w'_{\sigma(i)\sigma(k)} > 0$. For the last statement, J_0 depends on d_{jl} only through terms multiplied by w'_{jl} and, under persistent identities, the zero-value coupling is the identity, for which only pairs with $w_{ik}w'_{ik} > 0$ enter; hence altering d_{ik} on a pair unobserved in either state leaves the conclusion $D_0(G, H) = 0$ unchanged. More generally, J_0 does not depend on d_{jl} when $w'_{jl} = 0$. □

Proof of Proposition 2. Exchanging the dummy pairs $(i, j) \leftrightarrow (k, l)$ and using $w_{ki}w'_{lj} = w_{ik}w'_{jl}$ gives $Q(P) = \frac{1}{2}\sum w_{ik}w'_{jl}\{(c_{ik} - d_{jl})^2 + (c_{ki} - d_{lj})^2\}p_{ij}p_{kl}$. For reals a, a', b, b' with $\bar{a} = (a + a')/2$, $\check{a} = (a - a')/2$ and likewise for b, $(a - b)^2 + (a' - b')^2 = 2(\bar{a} - \bar{b})^2 + 2(\check{a} - \check{b})^2$. With $a = c_{ik}$, $a' = c_{ki}$, $b = d_{jl}$, $b' = d_{lj}$ this yields $Q_{C,D} = Q_{C^S,D^S} + Q_{C^A,D^A}$, and both are weighted sums of squares. □

Proof of Proposition 3. Since $x \log x - x \geq -1$, $J_\varepsilon \geq -\varepsilon nm$; continuity and coercivity follow as above. If a local minimizer had $p_{ab} = 0$, the right derivative of $t \mapsto J_\varepsilon(P + te_{ab})$ at 0 would be $-\infty$ because of the term $\varepsilon \log t$ (the other terms have bounded derivatives from above), a contradiction; hence local minimizers are interior and stationary. Coercivity is uniform in $\varepsilon \in (0, 1]$, so all minimizers lie in a compact set $\mathcal{K}$ on which $|E| \leq c_{\mathcal{K}}$; then $|V_\varepsilon - D_0| \leq \varepsilon c_{\mathcal{K}}$, and a cluster point $\bar{P}$ of minimizers satisfies $J_0(\bar{P}) \leq \lim\{V_\varepsilon + \varepsilon c_{\mathcal{K}}\} = D_0$. □

Proof of Lemma 1. (a) $Q(P + \tau D) = Q(P) + \tau\{\langle T(P), D\rangle + \langle T(D), P\rangle\} + \tau^2\langle T(D), D\rangle$ exactly, and $\langle T(D), P\rangle = \langle D, T^*(P)\rangle$ with $T^*(P)_{kl} = \sum_{ij} L_{ijkl}p_{ij}$. (b) Symmetric inputs give $L_{klij} = L_{ijkl}$. For $n = m = 2$, unit weights, $D = 0$ and $C = \left(\begin{smallmatrix} 0 & 1 \\ 0 & 0 \end{smallmatrix}\right)$, $T(P)_{1j} = r_2$ while $T^*(P)_{1j} = 0$. (c) Expand $(c_{ik} - d_{jl})^2$ and collect terms. □

Proof of Proposition 4. For $X \geq 0$ let $h_P(X) = \langle \nabla f(P), X\rangle + g(X)$. It is strictly convex and coercive, and its unique minimizer $Q = Q(P)$ is strictly positive (as in Proposition 3), so $\nabla h_P(Q) = 0$.

(a) Because Q is an interior minimizer, $\mathrm{gap}(P) = h_P(P) - h_P(Q) = B_{h_P}(P, Q)$, the Bregman divergence of h_P, which equals the Bregman divergence of g because the linear part does not contribute. The Bregman divergences of the two KL terms are non-negative by convexity, and that of εE is $\varepsilon\sum_{ij}\{p_{ij}\log(p_{ij}/q_{ij}) - p_{ij} + q_{ij}\} = \varepsilon\,\mathrm{KL}(P|Q)$. Hence

$\mathrm{gap}(P) \geq \varepsilon\,\mathrm{KL}(P|Q) \geq 0$. The gap vanishes if and only if $P = Q$, that is, if and only if P minimizes h_P, which for interior P is equivalent to $\nabla f(P) + \nabla g(P) = \nabla J_\varepsilon(P) = 0$.

Residual bound. Suppose all entries of P and Q lie in $[p_{\min}, p_{\max}]$ and their row and column sums are at least $r_{\min}$ and $c_{\min}$. On the segment $[P, Q]$ the Hessian of εE is at least $(\varepsilon/p_{\max})I$, so $\mathrm{gap}(P) \geq (\varepsilon/2p_{\max})\|P-Q\|_F^2$. Since $\nabla f(P)+\nabla g(Q) = 0$, $\nabla J_\varepsilon(P) = \nabla g(P)-\nabla g(Q)$, and the entrywise mean value theorem gives $\|\nabla g(P) - \nabla g(Q)\|_F \leq L_g\|P - Q\|_F$ with $L_g = \varepsilon/p_{\min} + \gamma m/r_{\min} + \gamma n/c_{\min}$. Therefore $\|\nabla J_\varepsilon(P)\|_F \leq L_g\{2p_{\max}\mathrm{gap}(P)/\varepsilon\}^{1/2}$. Because entropic couplings have entries of order $\exp(-\mathrm{cost}/\varepsilon)$, $p_{\min}$ is typically tiny and the bound is uninformative; this is why residuals are reported separately.

(b) Along $D = Q - P$, $f(P + \tau D) = f(P) + \tau\langle\nabla f(P), D\rangle + \tau^2 a$ exactly (Lemma 1), with $a = \alpha Q(D)$. Convexity of g gives $g(P + \tau D) \leq (1 - \tau)g(P) + \tau g(Q)$ for $\tau \in [0, 1]$, hence $J_\varepsilon(P + \tau D) \leq U(\tau) = J_\varepsilon(P) - \tau\,\mathrm{gap}(P) + a\tau^2$. If $a \leq 0$, $\tau_{\mathrm{ub}} = 1$ and $U(1) \leq J_\varepsilon(P) - \mathrm{gap}(P)$. If $a > 0$ and $\mathrm{gap}/(2a) \leq 1$, $U(\tau_{\mathrm{ub}}) = J_\varepsilon(P) - \mathrm{gap}^2/(4a)$; if $\mathrm{gap}/(2a) > 1$, $U(1) = J_\varepsilon(P) - \mathrm{gap} + a < J_\varepsilon(P) - \mathrm{gap}/2$. The accepted step satisfies $J_\varepsilon(P^{(r+1)}) \leq J_\varepsilon(P^{(r)} + \tau_{\mathrm{ub}}D) \leq U(\tau_{\mathrm{ub}})$ by construction, which gives the stated decrease. No property of the refined search is used.

(c) By (b) the iterates stay in the compact sublevel set $\mathcal{S}$. The oracle map $P \mapsto Q(P)$ is continuous (minimizer of a strictly convex coercive function depending continuously on its linear part), so $D^{(r)}$ is bounded on $\mathcal{S}$, and $a_r \leq \alpha \max_{ijkl} L_{ijkl}\|D^{(r)}\|_1^2 \leq A < \infty$. Summing (b) gives $\sum_r \min\{\mathrm{gap}_r/2, \mathrm{gap}_r^2/(4A)\} \leq J_\varepsilon(P^{(0)}) + \varepsilon nm < \infty$, so $\mathrm{gap}_r \to 0$. The gap is continuous on the closed orthant, so every limit point $\bar{P}$ has $\mathrm{gap}(\bar{P}) = 0$, i.e. $\bar{P} = Q(\bar{P}) > 0$, and $\bar{P}$ is stationary. □

Proof of Proposition 5. By definition, $\widehat{V}_\varepsilon(G, H) = V_\varepsilon(G, H) + e_\varepsilon(G, H)$. Substituting this expression into the definition of $\widehat{\Delta}_\varepsilon$ and subtracting the definition of Δ_ε gives $\widehat{\Delta}_\varepsilon - \Delta_\varepsilon = e_\varepsilon(G, H) - \frac{1}{2}e_\varepsilon(G, G) - \frac{1}{2}e_\varepsilon(H, H)$. The bound follows from the triangle inequality and the non-negativity of the three optimization errors. □

Proof of Proposition 6 and the portfolio estimate. $\Phi(P, \theta) = \nabla_P J_\varepsilon(P; \theta)$ is C^∞ on the open orthant times the parameter set. At $(P^\star, \theta_0)$, $\Phi = 0$ and $\partial_P\Phi$ is non-singular, so the implicit function theorem gives a unique C^1 branch $P^\star(\theta)$, and $\nabla v = \partial_\theta J_\varepsilon + (\partial_\theta P^\star)^\top \nabla_P J_\varepsilon = \partial_\theta J_\varepsilon$. The partial derivatives follow by direct differentiation; for example c_{ab} enters Q only

through terms with $(i,k)=(a,b)$, so $\partial Q/\partial c_{ab} = 2w_{ab}\{c_{ab}(PW'P^\top)_{ab} - (P(W'\circ D)P^\top)_{ab}\}$, and $\partial\,\mathrm{KL}(r|\mu)/\partial\mu_i = 1 - r_i/\mu_i$. For the portfolio estimate, suppose that near θ_0 each candidate k of the portfolio follows a C^1 branch with value $v_k(\theta)$, so that $\widehat{V}_\varepsilon = \min_k v_k$. If the minimum is attained by a unique $k^\star$ at θ_0, continuity gives $\widehat{V}_\varepsilon = v_{k^\star}$ in a neighbourhood and the formulas are its gradient. If several candidates attain the minimum, $\widehat{V}_\varepsilon$ is the minimum of finitely many C^1 functions and has directional derivatives $\min_{k\in\mathcal{K}^\star} \nabla v_k(\theta_0)^\top h$ along h, where $\mathcal{K}^\star$ is the set of minimizing candidates; it need not be differentiable. □

Proof of Proposition 7. (a) Delannoy numbers satisfy the same recursion as the number of paths. (b) $e^{-c^\star/\beta} \le \sum_A e^{-c_A/\beta} \le |\mathcal{A}_{T,S}| e^{-c^\star/\beta}$ with $c_A = \langle A, \Delta\rangle$ and $c^\star = \min_A c_A$. (c) Differentiating gives $\mathbb{E}_\beta[A_{ts}]$; every path contains $(1,1)$ and (T,S). (d) Transposition is a cost-preserving bijection between $\mathcal{A}_{T,S}$ and $\mathcal{A}_{S,T}$. (e) The diagonal path has cost 0 and other paths exist when $T \ge 2$. □

Proof of Proposition 8. By the definition of soft-DTW,
$\mathcal{A}_\beta(\Delta^{GH}) = -\beta \log \sum_{A\in\mathcal{A}_{T,S}} \exp\left\{-\frac{\langle A,\Delta^{GH}\rangle}{\beta}\right\}$. Since $\exp\left\{-\frac{\langle A,\Delta^{GH}\rangle}{\beta}\right\} = \prod_{(t,s)\in A} k_\beta(G_t, H_s)$, we obtain $\mathcal{A}_\beta(\Delta^{GH}) = -\beta \log K_\beta(\mathcal{G},\mathcal{H})$. Substitution into Definition 2 gives the claimed logarithmic representation. Suppose now that K_β is positive semidefinite on the trajectories under consideration. The Cauchy–Schwarz inequality for positive-semidefinite kernels yields $K_\beta(\mathcal{G},\mathcal{H})^2 \le K_\beta(\mathcal{G},\mathcal{G})K_\beta(\mathcal{H},\mathcal{H})$. All three kernel values are strictly positive because the state kernel is strictly positive and the alignment-path sets are nonempty. Consequently, $0 < \frac{K_\beta(\mathcal{G},\mathcal{H})}{\sqrt{K_\beta(\mathcal{G},\mathcal{G})K_\beta(\mathcal{H},\mathcal{H})}} \le 1$. Taking minus β times the logarithm proves the result. □

Proof of Proposition 9. With $\alpha = 0$ and $M_{ij} = C^{\mathrm{sp}}_{ij}$, J_ε is the entropic unbalanced OT objective with ground cost C^{sp}. For the second statement, Proposition 1(c,d) gives $D_0(G,H) > 0 = D_0(G,G) = D_0(H,H)$, and Proposition 3 gives $\Delta_\varepsilon(G,H) \to D_0(G,H)$. With $\alpha = 0$ the objective does not involve the geometries, so states with equal masses and attributes have $\Delta_\varepsilon = 0$. □

Proof of Proposition 10. (a) By Proposition 7(d), the alignment matrix of $\Delta^\top$ is $E^\top$, which reverses the sign of the offsets. (b) $\langle A, \Delta + \kappa \mathbf{1}\mathbf{1}^\top\rangle = \langle A,\Delta\rangle + \kappa|A|$; as $\kappa\to\infty$ the Gibbs distribution concentrates on the paths of minimal length, which for $T = S$ is the diagonal only. (c) Let $T = S$ and let a path have h horizontal steps; it then has h vertical

and $T-1-h$ diagonal steps and visits $T+h$ cells, so $h = |A|-T$. Index the visited cells by $k = 0, \dots, T+h-1$ and let $o_k = s-t$ at cell k. Since $o_0 = o_{T+h-1} = 0$ and each step changes the offset by $+1$, -1 or 0, with at most h steps of each sign, $|o_k| \le \min\{k, h, T+h-1-k\}$. Summing, $|\sum_k o_k| \le \sum_k \min\{k, h, T+h-1-k\} = h(h+1)/2 + h(T-1-h) + h(h-1)/2 = h(T-1)$, with equality for the path that takes all horizontal steps first and all vertical steps last. Hence $|\sum_{(t,s)\in A}(s-t)| \le (T-1)(|A|-T)$ for every path. Taking Gibbs expectations, $|\sum_{ts}(s-t)E_{ts}| \le (T-1)(\mathbb{E}_\beta|A|-T)$, and dividing by $\sum_{ts} E_{ts} = \mathbb{E}_\beta|A|$ gives the bound. (d) A cell has cost κ unless its offset equals ℓ. Since offsets start and end at 0 and change by at most one per step, a path that visits the shifted diagonal visits each offset $0, \dots, \ell-1$ at least once before reaching it and at least once after leaving it, so it has at least 2ℓ costly cells. The path with ℓ horizontal steps, then $T-1-\ell$ diagonal steps, then ℓ vertical steps has exactly 2ℓ costly cells; any other path visiting the shifted diagonal has a diagonal step at an offset different from ℓ, an offset above ℓ, or a repeated offset below ℓ, each of which adds a costly cell. A path that does not visit the shifted diagonal has all its $|A| \ge T$ cells costly, with equality only for the diagonal path. Therefore the minimal cost is $\kappa \min\{2\ell, T\}$: if $2\ell < T$ the shifted path is the unique minimizer, and if $2\ell > T$ the diagonal is. As $\beta/\kappa \to 0$ the Gibbs distribution concentrates on the unique minimizer, E converges to its indicator matrix, and τ converges to its average offset, $\ell(T-1)/(T+\ell)$ by the computation in (c), or to 0 for the diagonal. □

Exact decompositions. $\Delta_\varepsilon(G, H) = \sum_k \{J^k(P^\star_{GH}) - \frac{1}{2}J^k(P^\star_{GG}) - \frac{1}{2}J^k(P^\star_{HH})\}$ over the four terms k of J_ε, evaluated at the selected couplings. The two-player Shapley values satisfy $\phi_{\text{geo}} + \phi_{\text{node}} = v(\text{both})$ because $v(\emptyset) = \Delta_\varepsilon(G_t, G_t) = 0$.

S2 Algorithmic details

Oracle. $Q_{ij} = \exp\{(\phi_i + \psi_j - K_{ij})/\varepsilon\}$ with $\phi_i \leftarrow \lambda\varepsilon[\log \mu_i - \log \sum_j \exp\{(\psi_j - K_{ij})/\varepsilon\}]$, the analogous update for ψ, and $\lambda = \gamma/(\gamma+\varepsilon)$ (Chizat et al., 2018); iterations stop when the sup-norm change of the potentials is below 10^{-12}, with potentials warm-started across iterations. The contraction factor per sweep is of order λ^2, which explains the slow convergence in the near-balanced regime ($\gamma \gg \varepsilon$).

Safeguarded line search. With $\varphi(\tau) = J_\varepsilon(P + \tau D) - J_\varepsilon(P) = b\tau + a\tau^2 + g(P + \tau D) - g(P)$, each evaluation of φ or φ' costs $O(nm)$ because row and column sums of $P + \tau D$ are affine in τ. The majorant step is $\tau_{\mathrm{ub}} = 1$ if $a \le 0$ and $\min\{1, \mathrm{gap}/(2a)\}$ otherwise. If $a \ge 0$, φ is convex and the refined step is obtained by 60 bisection steps on φ'; if $a < 0$, $\varphi'' = 2a + g''$ with g'' convex in τ, so φ can have two local minima on $[0, 1]$, and the refined step is obtained from a 33-point grid and 60 golden-section steps. The refined step is accepted only if $\varphi(\tau_{\mathrm{ref}}) \le \varphi(\tau_{\mathrm{ub}})$.

Stopping rule. Algorithm 1 stops when gap $\le 10^{-9} \max\{1, |J_\varepsilon|\}$ (10^{-12}–10^{-15} in the verification experiments) or after 3000 iterations; the gap criterion was met in all reported runs.

Initializations. *Identity*: $\mathrm{diag}(\sqrt{\mu \circ \nu}) + 10^{-12}$ (same node sets). *Product*: $\mu\nu^\top / \sqrt{\mu^\top \mathbf{1}\, \nu^\top \mathbf{1}}$. *Attribute*: the exact solution of the convex $\alpha = 0$ problem. *Quantile*: north-west-corner coupling of the normalized masses after sorting the nodes of each state by their first attribute. *Annealing*: Algorithm 1 for $\varepsilon \in \{1, 0.5, 0.2, 0.1\}$ above the target and then at the target, each warm-started at the previous solution. *Random* (diagnostics only): $Z \circ \mu\nu^\top$ rescaled, with Z log-normal and drawn from a generator seeded by hashes of the two states, generated in a canonical orientation and transposed when the states are exchanged, so that the candidates are equivariant. Same-node comparisons use the identity, product and attribute starts; different node sets use product, attribute, quantile and annealing.

Software and hardware. Package `gsta` (Python 3.12.3, NumPy 2.4.4, SciPy 1.17.1, numba 0.68.0; POT 0.9.7 for cross-checks), double precision, one core of an Intel Xeon processor at 2.1 GHz with single-threaded OpenBLAS 0.3.31.

S3 Additional verification, audit and benchmark results

Gradients. The factorized gradient $T + T^*$ was compared with the explicit tensor formula $\sum_{kl}(L_{ijkl} + L_{klij})p_{kl}$ and with central differences of $Q(P)$ on nine random instances (unmasked 5×5, masked 5×5, masked 6×4); the maximum relative differences were

4.8×10^{-16} and 2.3×10^{-9}, while $2T$ differed by 39%–68%. Table S1 compares the envelope gradients of Proposition 5 with central differences (step 10^{-5}) in which Algorithm 1 is re-run at each perturbed input.

Table S1: Envelope gradients versus re-solved central differences.

Instance	Block	max. abs. error	max. rel. error
directed	C	1.4×10^{-9}	9.7×10^{-8}
directed	D	7.8×10^{-10}	4.0×10^{-8}
directed	F	4.2×10^{-9}	5.0×10^{-8}
directed	G	5.0×10^{-9}	6.0×10^{-8}
directed	μ	2.0×10^{-8}	9.5×10^{-8}
directed	ν	3.1×10^{-8}	1.2×10^{-7}
directed	C, with $rc^\top$ (incorrect)	8.9×10^{-3}	6.0×10^{-1}
directed+mask	C	1.6×10^{-9}	9.7×10^{-8}
directed+mask	D	1.6×10^{-9}	1.1×10^{-7}
directed+mask	F	6.2×10^{-9}	1.3×10^{-7}
directed+mask	G	6.4×10^{-9}	1.4×10^{-7}
directed+mask	μ	5.4×10^{-8}	1.4×10^{-7}
directed+mask	ν	4.4×10^{-8}	1.5×10^{-7}
different sizes	C	1.3×10^{-9}	7.2×10^{-8}
different sizes	D	1.8×10^{-9}	7.9×10^{-8}
different sizes	F	2.8×10^{-9}	2.1×10^{-8}
different sizes	G	2.3×10^{-9}	1.3×10^{-8}
different sizes	μ	4.1×10^{-8}	4.7×10^{-8}
different sizes	ν	3.1×10^{-8}	1.5×10^{-7}

Lead–lag index by exhaustive enumeration. For $T = S \leq 6$ all monotone paths (up to 1,683) were enumerated and τ was computed from the exact Gibbs probabilities for the idealized shift of Proposition 9(d) with $\kappa = 1$ and $\beta \in \{0.01, \ldots, 2\}$; for $T \in \{10, 20, 40\}$ the soft-DTW recursion was used. The enumeration agrees with the soft-DTW gradient to 4.1×10^{-15}. Table S2 shows that for $2\ell < T$ and $\beta/\kappa \leq 0.1$ the index equals the zero-temperature value $\ell(T-1)/(T+\ell)$ to at least four decimals, that it is further attenuated for larger β/κ, and that it collapses to zero when $2\ell > T$ (with ties between the two candidate

paths at $2\ell = T$). The pathwise bound of Proposition 9(c) was never violated in 2,000 random cost matrices.

Table S2: Lead–lag index for the idealized shift: finite-temperature values and the zero-temperature value.

T	ℓ	$\beta/\kappa = 0.05$	0.1	0.2	0.5	1	$\ell(T-1)/(T+\ell)$
6	0	0.0000	0.0000	0.0000	-0.0000	0.0000	0.0000
	1	0.7143	0.7143	0.7145	0.7008	0.5901	0.7143
	2	1.2500	1.2500	1.2486	1.1390	0.6980	1.2500
	3	1.0000	0.9995	0.9354	0.4305	0.1991	1.6667
	4	0.0000	0.0000	0.0001	0.0102	0.0149	2.0000
	5	0.0000	0.0000	0.0000	0.0001	0.0004	2.2727
10	1	0.8182	0.8182	0.8187	0.8161	–	0.8182
	2	1.5000	1.5000	1.5005	1.4815	–	1.5000
	3	2.0769	2.0769	2.0763	2.0121	–	2.0769
	5	1.8000	1.7973	1.4608	0.2206	–	3.0000
40	1	0.9512	0.9512	0.9515	0.9527	–	0.9512
	2	1.8571	1.8571	1.8578	1.8609	–	1.8571
	3	2.7209	2.7209	2.7221	2.7267	–	2.7209
	5	4.3333	4.3333	4.3355	4.3380	–	4.3333

Convergence audit. Table S3 summarizes 180 runs of Algorithm 1 (15 random instances of each type, product and random initializations, line-search grids of 33 and 513 points), with every iteration recorded. All runs converged, decreased the objective monotonically and satisfied the sufficient-decrease inequality of Proposition 4(b) at every step. Changing the grid changed the stationary point reached in 24 of 90 paired runs, all from random initializations, with the finer grid better in 11; from the product initialization the final value never changed.

Portfolio diagnostics. Table S4 reports, for twelve comparisons between networks with 8 and 6 nodes, the portfolio estimate, the estimate after adding 20 random initializations, the number of distinct stationary values among the 24 candidates, and the number of changes of the selected candidate under five perturbations of the geometry by relative noise 10^{-6}.

Table S3: Convergence audit of Algorithm 1: median [maximum] iterations, median shares of negative-curvature steps and of accepted refined steps, median residual $\|\nabla J_\varepsilon\|_\infty$ at termination, median time.

Instance type	runs	iterations	neg. curvature	refined accepted	residual	time (s)
different node sets	60	23 [80]	0.93	1.00	1.5×10^{-6}	0.010
same-node directed	60	27 [128]	0.94	0.93	1.6×10^{-6}	0.022
same-node masked	60	31 [76]	0.96	0.94	1.4×10^{-6}	0.022

Table S4: Multi-start diagnostics for difficult cross-network comparisons.

pair	winner (portfolio)	value	winner (+20 random)	rel. improvement	distinct values	switches
0	feature	0.7269	feature	0.0000	6	0
1	anneal	0.6912	anneal	0.0000	13	0
2	product	0.9334	random15	0.0000	5	0
3	anneal	0.5565	random9	0.0000	9	0
4	anneal	0.7229	random19	0.0028	9	0
5	anneal	0.6342	random13	0.0252	7	0
6	feature	0.8347	random13	0.0594	5	0
7	anneal	0.5546	random13	0.0000	12	0
8	anneal	0.7338	random18	0.0000	7	0
9	product	0.8790	random16	0.0000	10	0
10	anneal	0.4509	random14	0.0000	14	0
11	product	0.6476	random14	0.0000	2	0

Benchmarks. Table S5 breaks down the run time of Algorithm 1 for masked problems (median of three runs), and Table S6 compares it with POT on balanced, fully observed directed problems with $\alpha = 0.5$ (median of three instances); "energy excess" is the relative excess of the balanced fused GW energy of the returned coupling over that of POT's conditional gradient solution.

S4 Simulation designs and additional results

Base networks. Latent positions $x_i \sim N_2(0, I)$ give $S_{ik} = \|x_i - x_k\| + 2$; direction effects $u_i \sim N(0, 1)$ give $A_{ik} = \kappa(u_i - u_k)$, $\kappa = 0.5$; $C^0 = S + A$ with zero diagonal; masses are log-normal with log-scale standard deviation 1, normalized to total one; attributes are

Table S5: Run time of Algorithm 1 by component (masked problems, $\alpha = 0.5$, $\gamma = 1$, $\varepsilon = 0.05$).

N	iterations	s/iteration	oracle	structural	line search	3-start portfolio (s)
25	12	0.0023	71%	3%	19%	0.08
50	8	0.0083	78%	3%	18%	0.22
100	6	0.0347	83%	4%	12%	0.68
200	5	0.1482	81%	7%	12%	2.29
400	4	0.6220	82%	11%	9%	9.02

Table S6: Comparison with POT on balanced, fully observed directed problems.

N	method	time (s)	energy excess	marginal error
25	POT CG (exact OT oracle)	0.001	0	0
	POT entropic ($\varepsilon = 0.01$)	0.497	6.5×10^{-4}	7.4×10^{-5}
	Algorithm 1 ($\gamma = 10^3$, $\varepsilon = 0.01$)	4.311	1.3×10^{-4}	4.6×10^{-6}
50	POT CG (exact OT oracle)	0.002	0	0
	POT entropic ($\varepsilon = 0.01$)	0.405	6.4×10^{-3}	2.5×10^{-5}
	Algorithm 1 ($\gamma = 10^3$, $\varepsilon = 0.01$)	10.752	5.7×10^{-3}	2.8×10^{-6}
100	POT CG (exact OT oracle)	0.007	0	0
	POT entropic ($\varepsilon = 0.01$)	0.730	1.6×10^{-2}	2.1×10^{-5}
	Algorithm 1 ($\gamma = 10^3$, $\varepsilon = 0.01$)	53.614	1.6×10^{-2}	2.0×10^{-6}

$f_i \sim N(0, 1)$. Observed states add $N(0, \sigma^2)$ noise to every off-diagonal relation (truncated at zero), $N(0, 0.05^2)$ to attributes and a log-normal perturbation with standard deviation 0.02 to masses.

Detection. $n = 12$, periods $1, \ldots, 8$, $\sigma = 0.25$, shock in period 5 on the four heaviest nodes $\mathcal{H}$. Reversal: $c_{ik} = S_{ik} + (1 - 2\delta)A_{ik}$ on corridors within $\mathcal{H}$. Corridor surge: $c_{ik} = (1 + \delta)C^0_{ik}$ on the same corridors. Attribute shift: $f_i + 2\delta$ for $i \in \mathcal{H}$. Mass contraction: $\mu_i(1 - \delta/2)$ for $i \in \mathcal{H}$. The UFGW-based methods use the identity initialization when identities persist and the attribute and product initializations under relabeling. Variants change one factor at a time: $n = 24$ (still four shocked nodes), $\sigma = 0.4$, $\kappa = 1$, log-mass standard deviation 0.5, or an independent uniform relabeling of the nodes in every period.

Table S7 reports the base design with Wilson 95% intervals.

Table S7: Detection rates in the base design with Wilson 95% intervals (60 replications).

Scenario	δ	directed	symm.	α=0	w. Frob.	Frob.	summ.	spectr.
no shock	0	0.20 [0.12,0.32]	0.12 [0.06,0.22]	0.08 [0.04,0.18]	0.20 [0.12,0.32]	0.22 [0.13,0.34]	0.13 [0.07,0.24]	0.22 [0.13,0.34]
reversal	0.25	0.67 [0.54,0.77]	0.08 [0.04,0.18]	0.20 [0.12,0.32]	0.65 [0.52,0.76]	0.52 [0.39,0.64]	0.37 [0.26,0.49]	0.13 [0.07,0.24]
	0.5	0.92 [0.82,0.96]	0.08 [0.04,0.18]	0.10 [0.05,0.20]	0.92 [0.82,0.96]	0.78 [0.66,0.87]	0.37 [0.26,0.49]	0.25 [0.16,0.37]
	1	1.00 [0.94,1.00]	0.13 [0.07,0.24]	0.12 [0.06,0.22]	1.00 [0.94,1.00]	0.95 [0.86,0.98]	0.15 [0.08,0.26]	0.65 [0.52,0.76]
corridor surge	0.025	0.17 [0.09,0.28]	0.20 [0.12,0.32]	0.17 [0.09,0.28]	0.18 [0.11,0.30]	0.15 [0.08,0.26]	0.12 [0.06,0.22]	0.13 [0.07,0.24]
	0.05	0.32 [0.21,0.44]	0.38 [0.27,0.51]	0.08 [0.04,0.18]	0.30 [0.20,0.43]	0.17 [0.09,0.28]	0.27 [0.17,0.39]	0.30 [0.20,0.43]
	0.1	0.85 [0.74,0.92]	0.97 [0.89,0.99]	0.15 [0.08,0.26]	0.85 [0.74,0.92]	0.47 [0.35,0.59]	0.57 [0.44,0.68]	0.58 [0.46,0.70]
connectivity	0.05	0.23 [0.14,0.35]	0.28 [0.19,0.41]	0.82 [0.70,0.89]	0.23 [0.14,0.35]	0.08 [0.04,0.18]	0.42 [0.30,0.54]	0.05 [0.02,0.14]
	0.1	0.77 [0.65,0.86]	0.82 [0.70,0.89]	1.00 [0.94,1.00]	0.75 [0.63,0.84]	0.28 [0.19,0.41]	0.60 [0.47,0.71]	0.18 [0.11,0.30]
	0.2	1.00 [0.94,1.00]	1.00 [0.94,1.00]	1.00 [0.94,1.00]	1.00 [0.94,1.00]	0.67 [0.54,0.77]	0.67 [0.54,0.77]	0.18 [0.11,0.30]
mass contraction	0.25	0.10 [0.05,0.20]	0.12 [0.06,0.22]	0.50 [0.38,0.62]	0.17 [0.09,0.28]	0.23 [0.14,0.35]	0.98 [0.91,1.00]	0.92 [0.82,0.96]
	0.5	0.20 [0.12,0.32]	0.57 [0.44,0.68]	1.00 [0.94,1.00]	0.78 [0.66,0.87]	0.83 [0.72,0.91]	1.00 [0.94,1.00]	1.00 [0.94,1.00]
	1	1.00 [0.94,1.00]	1.00 [0.94,1.00]	1.00 [0.94,1.00]	1.00 [0.94,1.00]	1.00 [0.94,1.00]	1.00 [0.94,1.00]	1.00 [0.94,1.00]

Lead–lag. A network with $n = 10$ is drawn; with $w = \max(1, T/6)$ and $c = (T-1)/2 - \ell/2$, the first trajectory follows the pulse $p(t) = \exp\{-(t-c)^2/(2w^2)\}$ and the second, an independent noisy realization of the same network, the pulse centred at $c + \ell$. Level scenario: $C_t = C^0\{1 + p(t)\}$, $\mu_t = \mu\{1 - 0.2p(t)\}$, $f_t = f - 0.5p(t)$. Spread scenario: $C_t = \bar{c} + \{1 + 0.8p(t)\}(C^0 - \bar{c})$, with $\bar{c}$ the mass-weighted mean of C^0. Noise $\sigma = 0.25$; 24 replications per cell; identity initialization. Table S8 reports the results for β equal to one tenth of the median of the level-adjusted matrix, and Figure S1 the sensitivity to β.

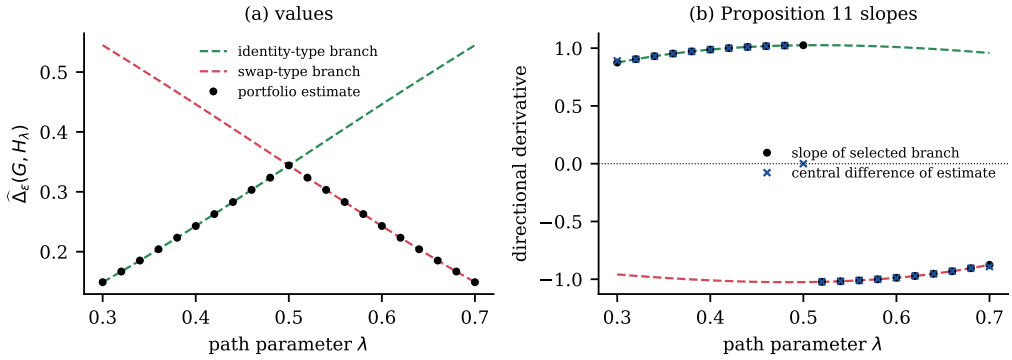


Figure S1: Mean lead–lag index as a function of β ($T = 10$).

Missing edges. $n = 20$, $T = 6$; $C_t = C^0 + tB + 0.25Z_t$ with B having $N(0, 0.15^2)$ entries. Designs: missing completely at random at rates 5%, 15%, 30%, 45%; asymmetric (probability 0.35 for pairs $i < j$ and 0.05 for $i > j$); cost-dependent (probability $0.5/[1 +$

$\exp\{-3(c_{ik} - q_{0.75})\}]$); node-dependent (probability 0.5 for relations from the five lightest nodes, 0.05 otherwise); structured (two destinations never observed, peripheral origins and costly relations more often missing). Thirty replications per design; identity initialization. Figure S2 shows means and standard deviations.

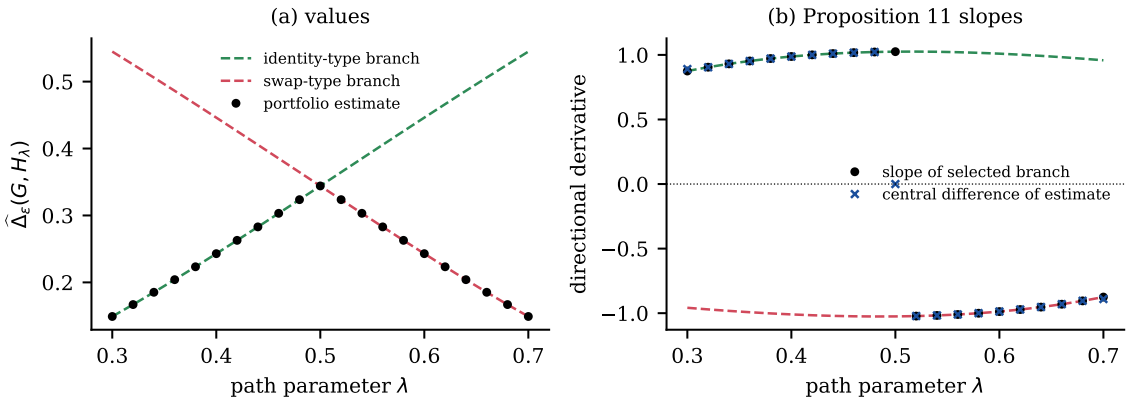


Figure S2: Relative error of the discrepancy matrix and Spearman correlation with the full-data matrix (mean ± standard deviation over 30 replications).

S5 Data and preprocessing

Sources. Three UNCTADstat datasets for 2016–2021, retrieved through the UNCTADstat API: transport costs of international trade (`US.TransportCosts`), filtered to transport mode `00` (all modes) and product `TOTAL`; the liner shipping connectivity index (`US.LSCI`), averaged over quarters within each year; and container port throughput (`US.ContPortThroughput`) in TEU. The exact extracts used for all results are included in the archive. Because the cost data cover all modes, the geometry is not restricted to maritime transport. Retrieval dates and the index base of the LSCI should be recorded from the retrieval logs.

Harmonization, coverage and exclusions. Economy labels were harmonized with the map {Korea, Republic of → Republic of Korea; Tanzania, United Republic of → United Republic of Tanzania; Congo, Dem. Rep. of the → Dem. Rep. of the Congo}. The intersection of the three files then contains 94 economies. The United States, the United Kingdom and France are absent from the transport-cost file, although they are present in the other two; they are therefore not part of the intersection. Eight economies with

incomplete throughput (Algeria, Dem. Rep. of the Congo, French Polynesia, Guinea, Iran, Libya, Montenegro, Mozambique) were excluded, leaving 86. Of the $86 \times 85 = 7{,}310$ directed pairs, 6,165 are observed in all six years and about 89% in any year. Coverage is directed: Mexico and Venezuela never appear as destinations, so their incoming costs never enter the comparison, while their outgoing costs do. One non-positive cost is treated as missing.

Preprocessing. Masses $\mu_{t,i} = \sqrt{\mathrm{TEU}_{t,i}}/\kappa$ with $\kappa = T^{-1}\sum_t \sum_i \sqrt{\mathrm{TEU}_{t,i}}$; attributes $f_{t,i} = (\log \mathrm{LSCI}_{t,i} - m)/s$ with pooled mean and standard deviation; geometry $c_{t,ij} = \min\{\max\{\log(1+\mathrm{cost}_{t,ij}), q_{0.005}\}, q_{0.995}\}/s_c$ with pooled quantiles and the pooled standard deviation s_c of the winsorized values; $c_{t,ii} = 0$; masks $w_{t,ij} = 1$ for observed pairs and $w_{t,ii} = 1$.

Directional asymmetry. For pairs observed in both directions, the table reports the mean of $|\log(r_{ij}/r_{ji})|$; its exponential, about 3.9, is the geometric mean of the ratio between the larger and the smaller directional cost, which is smaller than the arithmetic mean of that ratio.

An earlier preprocessing. An earlier version of this analysis set missing throughput to zero, which makes the mass of an economy collapse in the years in which it is not reported, imputed every missing cost by the maximum of its row, and used min–max scaling of $\log(1+\cdot)$ transforms. Under that preprocessing the imputed cells, about 13% of the entries, account for 82%–86% of the squared Frobenius norm of every geometry matrix, and the destination columns of Mexico and Venezuela are imputed entirely. The last row of Table 4 of the main text shows that, even then, the corrected solver does not single out 2020.

Regions. East Asia and Pacific (7): China, China Hong Kong SAR, Japan, Republic of Korea, Australia, New Zealand, Samoa. South and South-East Asia (9): Cambodia, India, Indonesia, Malaysia, Pakistan, Philippines, Singapore, Thailand, Viet Nam. Europe (25): Albania, Belgium, Bulgaria, Croatia, Cyprus, Denmark, Estonia, Finland, Germany, Greece, Iceland, Ireland, Italy, Latvia, Lithuania, Malta, Netherlands, Poland, Portugal, Romania, Russian Federation, Slovenia, Spain, Sweden, Ukraine. Middle East

and North Africa (13): Egypt, Israel, Jordan, Lebanon, Morocco, Oman, Qatar, Saudi Arabia, Sudan, Tunisia, Türkiye, United Arab Emirates, Yemen. Sub-Saharan Africa (11): Angola, Cameroon, Côte d'Ivoire, Ghana, Kenya, Mauritius, Namibia, Nigeria, Senegal, Togo, United Republic of Tanzania. Latin America and Caribbean (21): Argentina, Aruba, Brazil, Colombia, Dominican Republic, Ecuador, Grenada, Guatemala, Guyana, Honduras, Jamaica, Mexico, Nicaragua, Panama, Peru, Saint Lucia, Saint Vincent and the Grenadines, Suriname, Trinidad and Tobago, Uruguay, Venezuela.

S6 Additional empirical results

Table S10 reports the spectrum of $-\frac{1}{2}J\Delta J$ and the quality of the two-dimensional maps along the α path; no negative eigenvalues occur, so on these six states each Δ is a squared Euclidean distance matrix, but for $\alpha \geq 0.5$ two dimensions display only about half of the spectrum. Figure S3 shows the anomaly scores along the α path; the ordering of the years is the same for all $\alpha > 0$. Figure S4 displays the regional lead–lag indices of Table 5 of the main text, and Table S11 and Figure S5 the corridors with the largest first-order contributions to $\Delta(G_{2020}, G_{2021})$. For the regional comparisons, adding ten random initializations to the portfolio in each of the 540 cells improved no value, and the indices were unchanged; the smallest eigenvalues of the Gram matrices of k_β over the twelve states of each pair ranged from 0.027 to 0.77 and those of $k_\beta/(1 + k_\beta)$ from 0.005 to 0.31.

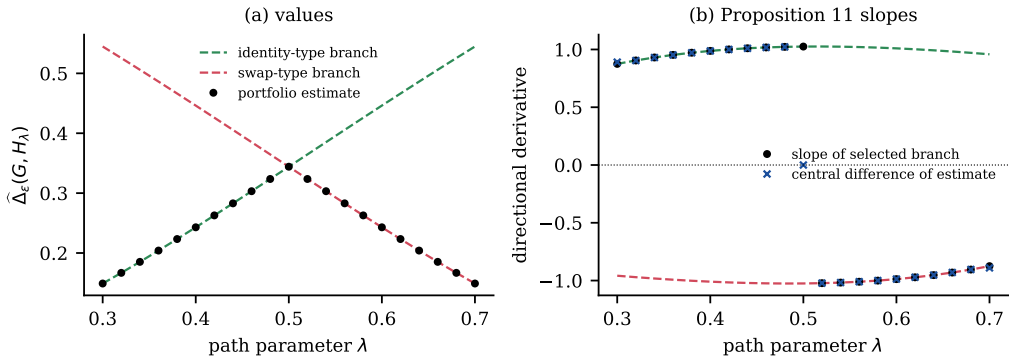


Figure S3: Anomaly and trend-adjusted scores, relative to the mean anomaly, along the α path.

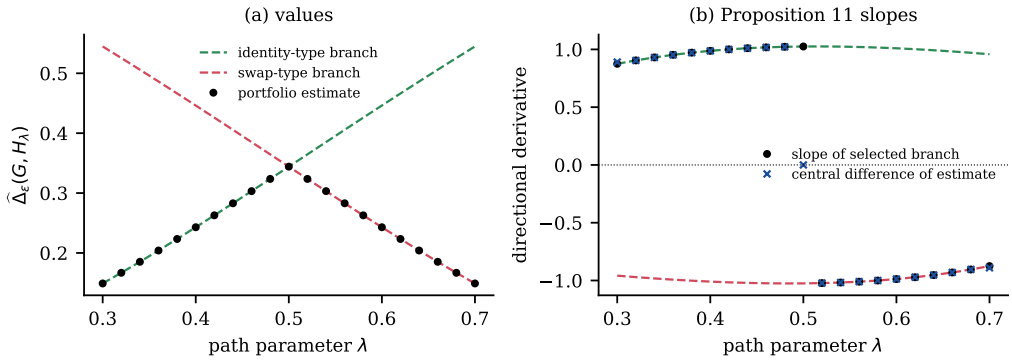


Figure S4: Level-adjusted regional lead–lag indices $\hat{\tau}(a, b)$ (row region a precedes column region b when positive), from directed and from symmetrized geometries.

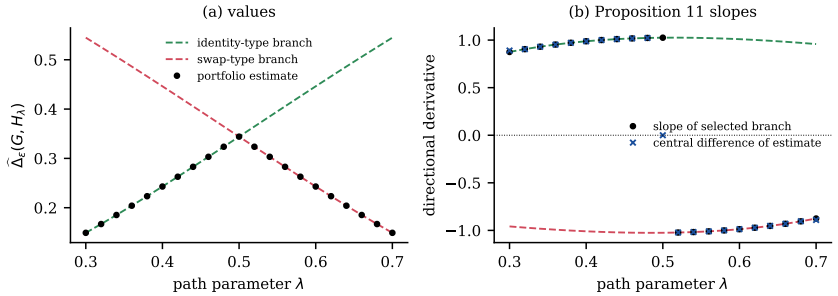


Figure S5: First-order corridor contributions to the 2020–2021 discrepancy.

S7 Reproducibility

The archive (version 1.1.0) contains the package `gsta/` (`ufgw.py`: Algorithm 1 with masks, safeguarded step, audit logging, residuals, portfolio and envelope gradients; `softdtw.py`: soft-DTW, gradients, Delannoy numbers, lead–lag index and boundary correction; `core.py`: trajectory-level quantities; `data.py`: construction of the panel; `sim.py`: simulation designs), the automated tests `tests/test_identities.py` (factorization and adjoint gradient against the explicit tensor, Proposition 2, stationarity, envelope gradient against re-solved differences, symmetry of the portfolio estimate, soft-DTW against enumeration, antisymmetry, pathwise bound, idealized shift, and zero G-STA between a trajectory and its copy), the experiment scripts, `make_tables.py` and `figures/make_figures.py`, which write every table and figure from the stored results, the processed data extracts, the retrieval scripts (credentials are read from environment variables), `requirements.txt` with the tested versions, and `scripts/run_all.sh`, which runs the tests and reproduces all results with one

command in about 1.5 hours on one core. All scripts use fixed seeds and are resumable. The sensitivity analysis adds `gsta_sensitivity.py` (Propositions 11–14, influence measures, budgeted perturbations, scaled-Hessian certificate), soft-DTW Hessian–vector products in `softdtw.py`, `verify_sensitivity.py` (Section S8.3), `policy_experiment.py`, whose stages reproduce Section 6.5 of the main text, and `make_outputs.py`, which writes its tables and figures.

S8 Sensitivity analysis

S8.1 Proofs

Proof of Proposition 11. (a) Under a regular selection the portfolio estimate coincides near the inputs with one branch value and is continuously differentiable with the gradient of Proposition 6 (proof of Proposition 6). The map $x \mapsto \widehat{V}_\varepsilon(G(x), G(x))$ composes $\widehat{V}_\varepsilon$ with the diagonal embedding, so its derivative is $\partial_{1,x} + \partial_{2,x}$; $\widehat{V}_\varepsilon(H, H)$ does not depend on x. (b) $J_\varepsilon^{GG}(P^\top) = J_\varepsilon^{GG}(P)$ (proof of Proposition 1(b)). The identity, product and attribute starts are symmetric when $G = H$. If P is symmetric, so is $\nabla f(P) = (1-\alpha)M + \alpha\{T(P) + T^*(P)\}$: M is symmetric, and renaming $(k, l) \to (l, k)$ in $T(P)_{ji} = \sum_{kl} w_{jk} w_{il} (c_{jk} - c_{il})^2 p_{kl}$ gives $T(P)_{ij}$, and likewise for T^*. The oracle problem with a symmetric cost and equal marginals is invariant under transposition and strictly convex, so its minimizer is symmetric, and the step is a scalar. Every iterate is therefore symmetric, and so is S. Substituting $S = S^\top$, $D = C$, $W' = W$ and $G = F$ in the formulas of Proposition 6 shows that $\partial_2 \widehat{V}_\varepsilon(G, G) = \partial_1 \widehat{V}_\varepsilon(G, G)$. Subtracting $\partial_1 \widehat{V}_\varepsilon(G, G)$ from $\partial_1 \widehat{V}_\varepsilon(G, H)$ gives the displayed formulas. (c) When H is a copy of G, the deterministic portfolio solves identical problems for (G, H) and (G, G), so $P = S$ and $r = r_S$, and the three expressions in (b) vanish. The raw gradient $\partial_{1,\mu} \widehat{V}_\varepsilon(G, G) = \gamma(\mathbf{1} - r_S/\mu)$ vanishes only if $r_S = \mu$. For a single node, the optimality condition $2\gamma \log(p/\mu) + \varepsilon \log p = 0$ gives $p = \mu^{2\gamma/(2\gamma+\varepsilon)} \neq \mu$ unless $\mu = 1$. Relations with $w_{ik} = 0$ enter neither problem, so their derivatives are zero (Corollary 1). □

Proof of Proposition 12. (a) $a(t)$ is linear in the entries Δ_{tu}, and $\Delta_{ut} = \Delta_{tu}$ because the portfolio estimate is symmetric. (b) $\mathcal{A}_\beta$ is smooth with gradient E (Proposition 7(c)). In $\Delta^{\mathcal{GH}}$ only row t depends on x. In $\Delta^{\mathcal{GG}}$ only row t and column t depend on x, except the entry (t, t), which is identically zero. $\Delta^{\mathcal{GG}}$ is symmetric, hence so is $E^{\mathcal{GG}}$ (Proposition 7(d)),

so the contributions of row and column t are equal and their sum cancels the factor $\frac{1}{2}$. (c) If $\mathcal{H}$ is a copy, then $\Delta^{\mathcal{GH}} = \Delta^{\mathcal{GG}}$ and $E^{\mathcal{GH}} = E^{\mathcal{GG}}$. For $s \neq t$ the problems (G_t, H_s) and (G_t, G_s) coincide, and $\nabla_x \Delta^{\mathcal{GH}}_{tt} = 0$ by Proposition 11(c), so the two sums cancel. For the temperature term, $\partial \mathcal{A}_\beta / \partial \beta = -\log \mathcal{Z} - \mathbb{E}_\beta \langle A, \Delta \rangle / \beta = \{\mathcal{A}_\beta - \langle E, \Delta \rangle\}/\beta$, where $\mathcal{Z}$ is the partition function; since $H(p_\beta) = \mathbb{E}_\beta \langle A, \Delta \rangle / \beta + \log \mathcal{Z}$, this equals $-H(p_\beta)$. □

Proof of Proposition 13. (a) $\mathcal{A}_\beta$ is a log-sum-exp of linear functions, so it is infinitely differentiable, and $Z \geq \max(T, S) > 0$. Differentiating the Gibbs expectation gives $\partial \mathbb{E}_\beta A_{ts} / \partial \Delta_{uv} = -\beta^{-1} \operatorname{Cov}_\beta(A_{ts}, A_{uv})$. From $\tau = \langle O, E \rangle / Z$ we get $d\tau = \langle O - \tau \mathbf{1}\mathbf{1}^\top, dE \rangle / Z = -(\beta Z)^{-1} \operatorname{Cov}_\beta(Y, \langle A, d\Delta \rangle)$. The bound follows from the Cauchy–Schwarz inequality and $\operatorname{Var}(A_{ts}) \leq \frac{1}{4}$. (b) Under the uniqueness conditions, near Δ the minimum is $\Delta_{k^\star}$ and the median of $\tilde{\Delta}$ is the fixed linear function $\langle \bar{e}_{\mathrm{med}}, \tilde{\Delta} \rangle$. Write $\hat{\tau}(\Delta) = \tau_1(\tilde{\Delta}/\beta)$, with τ_1 the index at unit temperature. The partial derivative with respect to β at fixed $\tilde{\Delta}$ is $-\langle g, \tilde{\Delta} \rangle / \beta$, and $\partial \beta / \partial \Delta = b(\bar{e}_{\mathrm{med}} - e_{k^\star})$ with $b/\beta = 1/\operatorname{med}(\tilde{\Delta})$; $\partial \tilde{\Delta} / \partial \Delta$ subtracts the derivative at $k^\star$. Collecting terms gives the formula. The adjusted matrix and the temperature are unchanged by $\Delta \mapsto \Delta + \kappa \mathbf{1}\mathbf{1}^\top$, and both scale with c under $\Delta \mapsto c\Delta$; differentiating $\hat{\tau}(\Delta + \kappa \mathbf{1}\mathbf{1}^\top) = \hat{\tau}(c\Delta) = \hat{\tau}(\Delta)$ at $\kappa = 0$ and $c = 1$ gives the two orthogonality relations. The minimum and the median are invariant under transposition, so Proposition 10(a) gives $\hat{\tau}(\Delta^\top) = -\hat{\tau}(\Delta)$; differentiating gives $\nabla \hat{\tau}(\Delta^\top) = -\nabla \hat{\tau}(\Delta)^\top$. (c) This follows from the chain rule, because only row t of $\Delta^{\mathcal{GH}}$ depends on x. □

Proof of Proposition 14. (a) $V_\varepsilon(G + \delta, H) \leq J_\varepsilon(P; G + \delta, H)$ because V_ε is a minimum, and Algorithm 1 started at P decreases the objective monotonically (Proposition 4(b)). For fixed P, J_ε is separable in C, F and μ, and the entropic term does not depend on them. Expanding $(c_{ik} + \delta c_{ik} - d_{jl})^2$ gives the linear term $\langle \partial_C J_\varepsilon, \delta C \rangle$ plus $\alpha \sum_{ik} w_{ik} \delta c_{ik}^2 \sum_{jl} w'_{jl} p_{ij} p_{kl}$, and the inner sum is $(PW'P^\top)_{ik}$. Expanding $\|f_i + \delta f_i - g_j\|^2$ adds $(1 - \alpha) \sum_i r_i \|\delta f_i\|^2$. Finally, $\mathrm{KL}(r|\mu + \delta\mu) - \mathrm{KL}(r|\mu) - \sum_i (1 - r_i/\mu_i)\delta\mu_i = \sum_i r_i \{x_i - \log(1 + x_i)\} \geq 0$ with $x_i = \delta\mu_i / \mu_i$. (b) Along the segment each branch $P^\star(\theta_\lambda)$ is continuously differentiable by the implicit function theorem. Hence $\nabla v(\theta) = \partial_\theta J_\varepsilon(P^\star(\theta), \theta)$ is continuously differentiable and v is twice continuously differentiable. $\mathcal{S}$ is a twice continuously differentiable function of finitely many branch values (linear combinations, soft-DTW, and $\hat{\tau}$ while its cells are fixed). The bound is Taylor's theorem with Lagrange remainder applied to $\lambda \mapsto \mathcal{S}(\theta_\lambda)$.

(c) Replacing u_k by $\pm\operatorname{sign}(\partial_k \mathcal{S}) v_k$ with $0 \le v_k \le \bar{u}_k$ cannot decrease the objective. The problem becomes the continuous knapsack $\max \sum_k |\partial_k \mathcal{S}| v_k$ subject to $\sum_k q_k v_k \le B$, which the greedy rule by ratio solves. $\square$

S8.2 Computational details

Given the selected couplings, each state-level gradient costs a few matrix products; all anomaly gradients of the application take seconds, whereas finite differences would need two solves for each of $6 \times 7{,}310$ corridors. Hessian–vector products of soft-DTW are obtained by differentiating the forward recursion for R and the backward recursion for E in a direction $\dot{\Delta}$. The gradient of $\hat{\tau}$ uses TS such products. Non-degeneracy is checked through the smallest eigenvalue of $\operatorname{diag}(P)^{1/2} \nabla_P^2 J_\varepsilon \operatorname{diag}(P)^{1/2}$, computed by Lanczos iterations with $\nabla_P^2 J_\varepsilon[X] = \alpha\{T(X) + T^*(X)\} + \gamma(X\mathbf{1}/r)\mathbf{1}^\top + \gamma\mathbf{1}(X^\top\mathbf{1}/c)^\top + \varepsilon X/P$. The congruence preserves definiteness and removes the ill-conditioning caused by small coupling entries. Semi-elasticities use $\partial c_{ik}/\partial \log \mathrm{cost}_{ik} = \mathrm{cost}_{ik}/\{(1 + \mathrm{cost}_{ik}) s_c\}$ for unwinsorized costs and zero otherwise, $\partial f_i/\partial \log \mathrm{LSCI}_i = 1/s_f$, and $\partial \mu_i/\partial \log \mathrm{TEU}_i = \mu_i/2$, with the pooled constants and κ fixed. Baseline gradients use a gap tolerance of 10^{-12} and counterfactual re-solves 10^{-10}. Every perturbed problem is solved with the portfolio and with Algorithm 1 started at the baseline coupling; a branch switch is recorded when the two values differ by more than $10^{-7} \max\{1, |J_\varepsilon|\}$. The cost-change benchmark of the counterfactuals moves, in order of the absolute observed change, each corridor towards its 2020 value without overshooting in (i) and away from its mean over the other years in (ii); it was not used in (iii), where the changes concern a whole trajectory.

S8.3 Verification

In the near copy the feature gradients are of order 10^{-6} and the absolute errors at most 7.4×10^{-11}; the large relative errors of the incorrect variants arise because the correct gradient nearly vanishes there.

The larger error for the masses of $\hat{\tau}$ reflects the step, which is not small relative to the regional masses (of order 10^{-2}), and the curvature of the index.

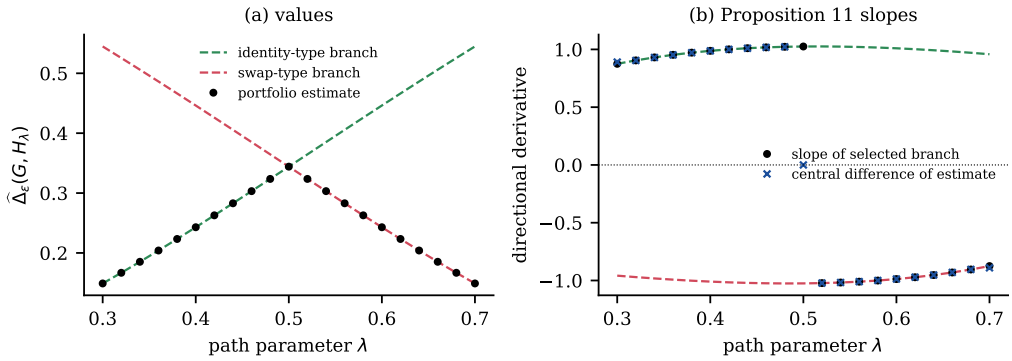


Figure S6: A branch crossing. H_λ interpolates the geometry of a six-node network and of a copy in which two interchangeable nodes exchange their relations. (a) Values of the two stationary branches (followed by warm starts) and the portfolio estimate (identity, product and attribute starts and six random starts). (b) Directional derivatives along the path from Proposition 11 for each branch, and central differences of the portfolio estimate; at $\lambda = 1/2$ the one-sided slopes are $+1.03$ and -1.03.

S8.4 Additional empirical results

Dropping 5% of the observed pairs at random in every year (four draws) left the anomaly ranking unchanged in every draw. The Spearman correlation of the corridor sensitivities with the baseline was at least 0.93 and that of the economy influence at least 0.975; at least 87% of the hundred most influential corridors and 90% of the ten most influential economies were retained in every year.

References

Chizat, L., Peyré, G., Schmitzer, B., and Vialard, F.-X. (2018). Scaling algorithms for unbalanced optimal transport problems. *Mathematics of Computation*, 87(314):2563–2609.

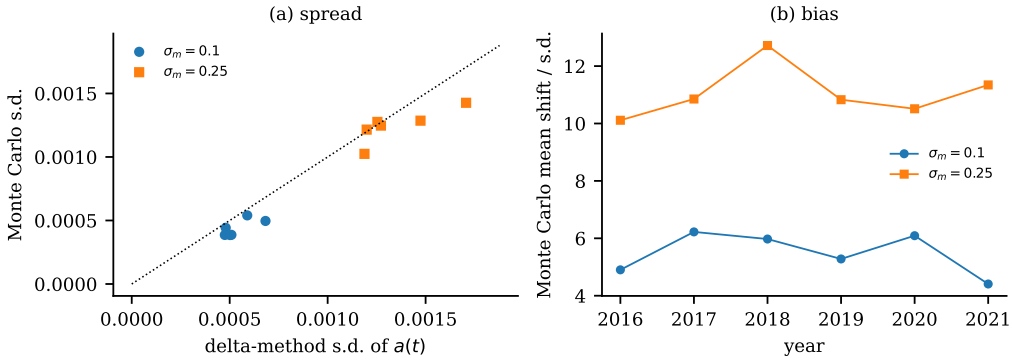


Figure S7: (a) Monte Carlo against delta-method standard deviations of the anomaly scores. (b) Monte Carlo mean shift of each score in units of its Monte Carlo standard deviation.

Table S8: Lead–lag estimation: mean index, mean boundary-corrected lead, its root mean squared error, and the share of replications with $\hat{\tau} > 0$ (24 replications). *: $2\ell \geq T$.

Shock	T	ℓ	mean $\hat{\tau}$	mean $\hat{\ell}$	RMSE $\hat{\ell}$	sign correct
level	6	0	0.00	0.00	0.02	–
		1	0.71	1.00	0.00	1.00
		2	1.31	2.12	0.13	1.00
		3*	1.66	2.99	0.03	1.00
		4*	-0.34	-0.45	4.45	0.00
	10	0	-0.00	-0.00	0.02	–
		1	0.81	0.99	0.01	1.00
		2	1.52	2.04	0.04	1.00
		3	2.09	3.03	0.03	1.00
		4	2.63	4.13	0.13	1.00
	16	0	-0.00	-0.00	0.03	–
		1	0.84	0.95	0.05	1.00
		2	1.65	1.97	0.05	1.00
		3	2.40	3.05	0.05	1.00
		4	3.06	4.10	0.10	1.00
spread	6	0	-0.00	-0.00	0.04	–
		1	0.71	0.99	0.03	1.00
		2	1.26	2.02	0.15	1.00
		3*	1.65	2.95	0.13	1.00
		4*	-0.34	-0.47	4.50	0.04
	10	0	-0.00	-0.00	0.09	–
		1	0.73	0.88	0.23	1.00
		2	1.43	1.90	0.21	1.00
		3	2.08	3.00	0.11	1.00
		4	2.56	3.97	0.16	1.00
	16	0	0.07	0.08	0.21	–
		1	0.76	0.86	0.33	1.00
		2	1.51	1.80	0.35	1.00
		3	2.23	2.80	0.29	1.00
		4	2.92	3.88	0.28	1.00

Table S9: Descriptive statistics of the 86-economy panel. *Median over the 6,165 pairs observed in all years. The last two columns use pairs observed in both directions.

Year	observed	median USD/kg*	TEU (m)	mean log LSCI	mean $\lvert\log(r_{ij}/r_{ji})\rvert$	corr$(\log r_{ij}, \log r_{ji})$
2016	0.890	0.195	583.6	4.599	1.43	0.16
2017	0.887	0.153	622.4	4.576	1.35	0.18
2018	0.894	0.151	654.3	4.586	1.33	0.17
2019	0.891	0.154	673.1	4.572	1.32	0.16
2020	0.889	0.161	670.4	4.564	1.36	0.16
2021	0.896	0.192	714.7	4.563	1.31	0.19

Table S10: Classical scaling diagnostics: leading eigenvalues of $-\frac{1}{2}J\Delta J$, share of the positive spectrum displayed by two dimensions, share of negative eigenvalues, and Kruskal stress of the two-dimensional embedding of $\Delta^{1/2}$.

α	eigenvalues of $-\frac{1}{2}J\Delta J$	first two / positive	negative / total	stress
0.00	0.005, 0.001, 0.000, 0.000, 0.000	0.88	0.00	0.12
0.25	0.043, 0.031, 0.021, 0.018, 0.015	0.58	0.00	0.36
0.50	0.086, 0.065, 0.048, 0.041, 0.034	0.55	0.00	0.37
0.75	0.114, 0.091, 0.068, 0.059, 0.049	0.54	0.00	0.37
1.00	0.124, 0.103, 0.080, 0.071, 0.059	0.52	0.00	0.40

Table S11: Corridors with the largest first-order contributions to $\Delta(G_{2020}, G_{2021})$; Δc_{ij} is the change of the standardized log cost.

Origin	Destination	$\partial\Delta/\partial c_{ij}$	Δc_{ij}	contribution
Malta	China, Hong Kong SAR	9.1×10^{-4}	+5.73	5.2×10^{-3}
Morocco	Cambodia	3.6×10^{-4}	+4.99	1.8×10^{-3}
Yemen	Brazil	3.4×10^{-4}	+5.23	1.8×10^{-3}
Cote d'Ivoire	Jamaica	2.3×10^{-4}	+6.06	1.4×10^{-3}
Panama	Oman	-4.7×10^{-4}	-2.98	1.4×10^{-3}
Panama	Lithuania	2.2×10^{-4}	+5.97	1.3×10^{-3}
Jamaica	China, Hong Kong SAR	4.4×10^{-4}	+3.01	1.3×10^{-3}
Dominican Republic	Nigeria	2.2×10^{-4}	+5.85	1.3×10^{-3}
China, Hong Kong SAR	Egypt	6.7×10^{-4}	+1.89	1.3×10^{-3}
Colombia	Malta	3.1×10^{-4}	+3.57	1.1×10^{-3}
Viet Nam	Panama	-6.5×10^{-4}	-1.69	1.1×10^{-3}
Lebanon	Panama	2.7×10^{-4}	+3.94	1.1×10^{-3}

Table S12: Proposition 11 versus central differences re-solved with the full portfolio (tolerance 10^{-13}, step 10^{-5}); 8 coordinates per block and argument; relative error $\|\text{analytic} - \text{FD}\|_\infty / \|\text{FD}\|_\infty$, maximum over both arguments. $\lambda_{\min}$: smallest eigenvalue of the scaled Hessian. Derivatives with respect to unobserved relations were zero in every instance.

			Proposition 11			incorrect variants	
Instance	gap	$\lambda_{\min}$	C	F	μ	self-argument omitted	raw va
directed	2.2×10^{-15}	0.050	7.9×10^{-8}	6.1×10^{-8}	2.2×10^{-7}	0.18	0
directed + 25% missing	5.2×10^{-14}	0.027	8.0×10^{-7}	4.6×10^{-7}	7.5×10^{-6}	0.25	0
8 vs. 6 nodes, 20% missing	5.8×10^{-14}	0.050	1.1×10^{-6}	1.3×10^{-6}	8.3×10^{-7}	0.06	0
near copy	$< 10^{-15}$	0.050	8.7×10^{-8}	9.1×10^{-5}	6.7×10^{-5}	118	

Table S13: Propositions 12 and 13 on two simulated trajectories of five states (8 and 6 nodes, 15% missing relations, common pulse with opposite timing; $\hat{\tau} = 0.355$) versus re-solved central differences (tolerance 10^{-13}, step 10^{-5}), with respect to the inputs of the third state of the first trajectory. $a(1)$: anomaly score of another state. G-STA at fixed β; $\hat{\tau}$ with the level adjustment and the data-dependent temperature. All 25 cross cells had regular selections.

Quantity	block	max. abs. error	rel. error	max. |FD|
$a(3)$	C	1.6×10^{-9}	3.8×10^{-7}	4.3×10^{-3}
	F	5.7×10^{-9}	1.1×10^{-7}	0.050
	μ	6.8×10^{-7}	1.5×10^{-6}	0.45
$a(1)$	C	4.6×10^{-10}	1.9×10^{-7}	2.4×10^{-3}
	F	1.9×10^{-9}	1.2×10^{-7}	0.016
	μ	2.7×10^{-8}	2.0×10^{-7}	0.13
$\widehat{\text{GSTA}}$	C	8.8×10^{-10}	3.0×10^{-7}	2.9×10^{-3}
	F	4.2×10^{-8}	8.9×10^{-7}	0.047
	μ	3.6×10^{-7}	3.5×10^{-7}	1.01
$\hat{\tau}$	C	1.8×10^{-8}	5.3×10^{-6}	3.4×10^{-3}
	F	1.1×10^{-7}	8.9×10^{-7}	0.12
	μ	3.2×10^{-6}	8.6×10^{-6}	0.37

Table S14: Gradients on the data versus central differences re-solved with the full portfolio (tolerance 10^{-12}, step 10^{-4}). Corridors: the largest gradients and randomly chosen ones. EAP–SSEA: inputs of the SSEA state of 2018.

Quantity	block	coordinates	max. abs. error	rel. error	max. |FD|
$\Delta(G_{2021}, G_{2020})$	C	16	2.3×10^{-9}	2.4×10^{-6}	9.3×10^{-4}
	F	4	1.8×10^{-9}	8.8×10^{-7}	2.0×10^{-3}
	μ	4	6.7×10^{-6}	1.5×10^{-5}	0.45
$a(2020)$	C	6	1.5×10^{-9}	1.4×10^{-6}	1.1×10^{-3}
$\widehat{\mathrm{GSTA}}$(EAP,SSEA)	C	10	1.6×10^{-9}	7.7×10^{-6}	2.1×10^{-4}
	F	3	6.6×10^{-9}	4.3×10^{-7}	0.015
	μ	3	2.7×10^{-5}	5.4×10^{-5}	0.50
$\hat{\tau}$(EAP,SSEA)	C	10	3.2×10^{-7}	9.2×10^{-6}	0.035
	F	3	1.7×10^{-6}	4.2×10^{-6}	0.41
	μ	3	8.9×10^{-3}	9.9×10^{-4}	9.0

Table S15: Most influential economies for the anomaly score: total corridor influence S_i on the log-cost scale, imbalance B_i, semi-elasticities with respect to log LSCI and log TEU, and mass μ_i.

Year	economy	$S_i \times 10^3$	B_i	$\partial a/\partial \log \mathrm{LSCI}_i$	$\partial a/\partial \log \mathrm{TEU}_i$	μ_i
2016	China, Hong Kong SAR	20.8	+0.00	1.3×10^{-3}	4.6×10^{-3}	0.027
	China	15.2	+0.32	-9.5×10^{-3}	-2.5×10^{-4}	0.087
	Singapore	13.4	−0.06	-1.6×10^{-3}	2.4×10^{-3}	0.034
	Japan	9.4	+0.27	1.1×10^{-3}	9.1×10^{-4}	0.029
	Philippines	8.9	+0.23	1.6×10^{-4}	1.9×10^{-3}	0.017
	Malaysia	8.9	+0.05	5.5×10^{-4}	7×10^{-5}	0.030
2020	China, Hong Kong SAR	11.4	+0.16	-1.5×10^{-3}	3.4×10^{-3}	0.026
	Singapore	6.6	+0.26	4.0×10^{-4}	1.6×10^{-3}	0.037
	China	6.3	−0.12	2.9×10^{-3}	2.1×10^{-3}	0.095
	Panama	5.7	−0.06	1.8×10^{-4}	2.1×10^{-3}	0.017
	Malta	5.6	+0.53	-1.3×10^{-3}	2.7×10^{-3}	0.010
	United Arab Emirates	5.6	−0.48	-8.5×10^{-4}	1.0×10^{-3}	0.027
2021	China, Hong Kong SAR	15.9	+0.06	-1.2×10^{-3}	4.2×10^{-3}	0.026
	China	15.3	+0.30	1.4×10^{-2}	3.7×10^{-3}	0.098
	Japan	8.6	+0.10	-1.2×10^{-3}	1.7×10^{-3}	0.029
	Singapore	8.2	+0.14	1.1×10^{-3}	2.2×10^{-3}	0.038
	Philippines	7.5	+0.49	-2.0×10^{-3}	2.3×10^{-3}	0.018
	Germany	6.7	−0.23	-1.9×10^{-3}	1.1×10^{-3}	0.024

Table S16: Corridors with the largest semi-elasticity $\partial a(t)/\partial \log \text{cost}_{ik}$, with the semi-elasticity of the reverse corridor.

Year	origin	destination	cost (USD/kg)	semi-elasticity	reverse
2020	Malta	China	5.24	1.9×10^{-3}	2.2×10^{-5}
	China, Hong Kong SAR	United Arab Emirates	3.24	-1.8×10^{-3}	6.2×10^{-4}
	Morocco	Ecuador	3.72	7.3×10^{-4}	9.2×10^{-7}
	Dominican Republic	Egypt	13.49	6.8×10^{-4}	-1.5×10^{-6}
	Colombia	Egypt	2.41	6.5×10^{-4}	-1.6×10^{-6}
	United Arab Emirates	China, Hong Kong SAR	4.23	6.2×10^{-4}	-1.8×10^{-3}
	Singapore	Morocco	1.16	5.6×10^{-4}	2.1×10^{-6}
	Ghana	China, Hong Kong SAR	7.50	5.4×10^{-4}	-8.6×10^{-6}
2021	Malta	China	6.34	1.7×10^{-3}	2.7×10^{-5}
	China, Hong Kong SAR	United Arab Emirates	3.46	-1.5×10^{-3}	2.5×10^{-4}
	Jamaica	China, Hong Kong SAR	10.51	9.8×10^{-4}	2.2×10^{-5}
	Malta	China, Hong Kong SAR	20.72	9.3×10^{-4}	9.4×10^{-7}
	Morocco	China, Hong Kong SAR	3.01	8.8×10^{-4}	-4.7×10^{-5}

Table S17: First-order predictions of the gradient strategy and re-solved changes. (i) $\Delta(G_{2020}, G_{2021})$, with the cross-term change, its first-order prediction and the frozen-coupling remainder R_P of Proposition 14(a). (ii) $a(2020)$. (iii) $\hat{\tau}(\mathrm{EAP}, \mathrm{SSEA})$. Portfolio and branch-following values agreed in every case.

	B	predicted	realized	ratio	cross-term change	cross-term prediction	R_P
(i)	1	-1.54×10^{-3}	-1.34×10^{-3}	0.87	-1.34×10^{-3}	-1.49×10^{-3}	2.0×10^{-4}
	4	-4.20×10^{-3}	-2.89×10^{-3}	0.69	-2.96×10^{-3}	-4.03×10^{-3}	1.2×10^{-3}
	16	-1.10×10^{-2}	-7.45×10^{-3}	0.68	-7.60×10^{-3}	-1.04×10^{-2}	3.2×10^{-3}
	64	-2.39×10^{-2}	-1.59×10^{-2}	0.66	-1.62×10^{-2}	-2.28×10^{-2}	7.7×10^{-3}
	128	-3.36×10^{-2}	-2.16×10^{-2}	0.64	-2.23×10^{-2}	-3.20×10^{-2}	1.1×10^{-2}
(ii)	8	5.3×10^{-3}	6.6×10^{-3}	1.25			
	32	1.17×10^{-2}	1.47×10^{-2}	1.26			
	128	2.20×10^{-2}	3.05×10^{-2}	1.39			
	512	3.30×10^{-2}	5.03×10^{-2}	1.52			
	2048	3.95×10^{-2}	6.30×10^{-2}	1.59			
(iii)	0.5	-0.014	-0.027	1.87			
	2	-0.043	-0.064	1.50			
	8	-0.101	-0.136	1.35			
	16	-0.146	-0.186	1.27			

Table S18: Delta-method and Monte Carlo standard deviations of the anomaly scores and of two margins under independent multiplicative cost errors with log-scale standard deviation σ_m (12 replications per level, all problems re-solved). Shift: Monte Carlo mean minus baseline. Rank: mean Monte Carlo rank.

	$\sigma_m = 0.10$				$\sigma_m = 0.25$			
	delta s.d.	MC s.d.	shift	rank	delta s.d.	MC s.d.	shift	rank
$a(2016)$	0.68	0.50	2.43	1.00	1.71	1.43	14.43	1.00
$a(2017)$	0.51	0.39	2.41	4.92	1.27	1.25	13.53	4.58
$a(2018)$	0.48	0.39	2.31	6.00	1.19	1.02	13.03	6.00
$a(2019)$	0.48	0.44	2.34	3.00	1.20	1.22	13.16	3.08
$a(2020)$	0.50	0.39	2.35	4.08	1.25	1.28	13.41	4.33
$a(2021)$	0.59	0.54	2.38	2.00	1.47	1.29	14.59	2.00
$a(2016) - a(2020)$	0.66	0.51	0.09	–	1.64	2.30	1.01	–
$a(2021) - a(2020)$	0.60	0.52	0.03	–	1.51	1.73	1.17	–

All entries except ranks $\times 10^{-3}$.